environments and providers—public and private clouds, on-premises data centres, and edge locations—to create a practically unlimited and adaptable cloud architecture. This concept is not new for those who have heard of Massachusetts Institute of Technology's SuperCloud. For a few years now, MIT has been supporting research projects that require significant computing, memory, or big data resources through its SuperCloud, a shared cluster of thousands of central processing units (CPUs) and graphics processing units (GPUs). Researchers get access to on-demand high-performance computing within seconds—a job using 320 billion could be launched in only 4 seconds, as the team demonstrated in 2018.

So, what has changed is that the kind of computing that was required by researchers some years ago is now required by innumerable enterprises that wish to implement AI on a large scale! As a result, we are seeing a surge in the availability of hybrid cloud computing solutions—from all the industry majors, such as IBM, Google, Microsoft, and Amazon.

This party on the clouds has become even more interesting with the entry of quantum computing! Readers following our stories on quantum computing will be aware of the slow and steady 'democratisation' that has been going on. By making quantum computing capabilities available on hybrid clouds, industry giants have enabled even small organisations and researchers to use quantum computing for research and innovation without heavy hardware investments or complex coding. 2025 will surely see a surge in the use of such quantum computing capabilities for mainstream applications.

On top of everything else, it is interesting to note that AI itself is now being used to manage and optimise the hybrid cloud infrastructure that it uses! AI is good at resource allocation and threat management, enabling greater performance and efficiency at lower cost points.

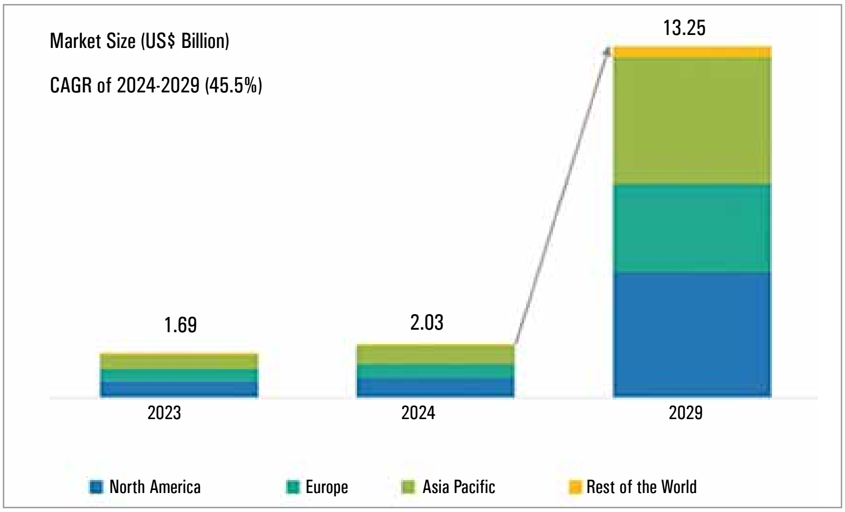

Figure 1: Growth in the humanoid robot market (Source: MarketsandMarkets)

Robotics and automation

Gone are the days when rare 'robot sightings' were the talk of the town! Here we are, in 2025, discussing the cost of adopting robots in various industries, without any doubt that they have arrived!

Robotics and automation, like every other tech sphere, are also expected to be greatly influenced by AI. While low-code, no-code development is expected to fuel digital automation, AI-powered robots are expected to make their presence felt in the physical world.

Those following the trends in robotics are sure to have bumped into the latest jargon—mobile manipulator or MoMa. Also known as a mobile cobot, MoMa lies at the intersection of collaborative robots (cobots) and mobile robots. It consists of a mobile base, a deft collaborative robotic arm, and some smartness—machine vision, sensors, and AI, to move around and work autonomously yet safely.

Specifically, experts are betting on the increased adoption of multifunctional or polyfunctional robots—those that can perform multiple tasks and seamlessly switch between them without the need for any physical restructuring. This gives all the economic benefits of reusability—such as easy deployment, improved efficiency, faster RoI, lower risk and scalability.

Nvidia experts predict that as agentic AI brings new intelligence to robots, volume will increase and costs will come down sharply. The average cost of industrial robots is expected to be around $10,800 in 2025. SLMs, which are energy-efficient and have lower latency, are expected to improve the functionality of robots operating at the edge greatly. "The shift to small language models in edge computing will improve inference in various industries, including automotive, retail, and advanced robotics. It also can train robots to react in virtually any scenario, enhancing their adaptability and performance across different clinical situations," says the Nvidia report.

It is also expected that AI, together with virtual reality (VR), will help to train robots to handle complex tasks such as surgical assistance. Virtual worlds will also help to train the robots that interact with humans, by simulating different scenarios. This will reduce the cost of training and the risk of deployment, paving the way for greater adoption of co-working robots.